



Attachment 1:	Job Description
Title of Position:	Senior Data analyst
Date of Job Description	01 April 2023
Reporting to:	Senior Technical Advisor
Job Summary: Reporting to the Senior Technical Advisor, the senior data analyst will provide technical leadership and direction on statistical analysis and research agenda for the PMA project. The incumbent will provide leadership in data collection, management, analysis, and utilization activities including additional analysis to support scientific writing. S/he will advise the project on statistical interpretation and implications of the results for program planning and decision-making. S/he will also support the technical capacity-building efforts of the project team. S/he will serve as an expert in advanced methods of statistical analysis, advanced programming consultation, planning, and conducting analyses of complex datasets.	
Key Responsibilities	
Survey design and operations: - Provides advice/technical support to the project team and stakeholders in regard to the statistical aspects of research and program evaluation such as study design, sample size determination, power analysis, data management, analysis of data, presentation of statistical information and other related issues. - Participate in the protocol development and advice on designing of data collection tools and data analysis and translation plan. - Participates in identification and selection of the technology platform for the program database system and then design, develop and implement relational database. - Provides technical support to project teams to understand indicators at the design level. - Assists in defining data to gather for surveillance, program evaluation and research; gathers and reviews current statistical literature; assists in defining population of interest and sample sizes for specific groups for surveillance, program evaluation and research activities, and develops statistical summary reports based on those activities. - Working with the programmer, data manager and demographer to identify data production process improvement, opportunities and propose system modifications on devise data governance strategies. - Prepare the cleaning do files to be used for data cleaning, and perform analysis to assess the quality and meaning of during data collection to ensure that data of high quality are generated. - Takes a lead role in generating and managing survey weights in consultation with KNBS Technical team.	
Statistical design and analysis - Use statistical tools to identify, analyze and interpret patterns and trends emerging from the project data. - Analyze large-scale data sets such as National Censuses and Demographic and Health Surveys to support data triangulation with PMA data.	



- Interprets and analyzes data for the project team and assists them in the interpretation of the statistical implications of the data for public health policy and program activities;
- Prepare final analysis reports for the stakeholders to understand the data analysis steps, enabling them to make program and policy decisions based on survey findings.
- Support computing the indicators to allow SDG reporting using PMA data.
- Facilitate preparing and reviewing project data and related materials for public release.
- Assist the supervisor to plan and oversee M&E activities ensuring effective use of data in decision-making.
- Apply and promote statistical work based on **best practices** and evidence-based principles.

Data utilization & Communication of survey findings:

- Collaborate with team members to write protocols, reports, abstracts and manuscripts and to communicate statistical results in meetings and conferences locally and internationally.
- Show a high level of attention to detail during the compilation of large amounts of statistical output to ensure accurate presentation of and reporting of results.
- Takes a lead role in data visualization and communication by ensuring that all the data products are designed and tailored to respond to the needs of specific audiences.
- Communicate statistical information to non-statisticians both in written and spoken form.
- Ensure that data including confidential records in areas of responsibility are maintained appropriately.
- Support university students and faculty members utilizing PMA data to understand PMA indicators at the design level.
- Represent PMA project at local, National and other consultative stakeholder meetings.
- Participate in using current research work to support business development in the organization.

Research agenda and manuscript development

- Participate, take lead role in PMA publication process, and explore with support from the Principal Investigator to identify possible papers to be generated using the project data.
- Assist in selection of statistical methods for data analysis, in authoring and reviewing the corresponding sections of the program evaluation and research protocols and methods sections of reports, abstracts and manuscripts, and in conducting the analysis once datasets are created.
- Create social bridges that allow for open sharing and re-use of PMA data among stakeholders including university faculty members and students.
- Train and support university staff and students in accessing PMA data for research purposes.
- Assist the project PI and Co-PI in providing technical support to master's and PhD students using PMA data for academic research – selection of statistical methods for data analysis, review of corresponding sections of their research reports/publications/theses.

Training and capacity building

- Provide statistical education and training to the project staff.
- Guide/supervise Data management team in all statistical matters, including statistical methods, sample size planning, data entry, and development of data analysis plans, data

analysis, presentation of the results and review of protocols, reports and manuscripts.

- Provide statistical expertise for seminars and strategy discussions.
- Provide statistical technical assistance to data management staff required to provide cleaned and verified datasets for analysis, including efficient scoring, processing and extraction of datasets
- Participate in key data management and analysis training both locally and internationally
- Actively pursue and participate in self-development activities in consultation with the supervisor e.g., attending advanced training courses and seminars.
- Provide leadership and work effectively as a member of the project team

Requirements

- An MSC in Applied Statistics or Bio-statistics or a quantitative social science discipline (e.g., Demography, Economics).
- Minimum of seven (7) years of relevant hands-on experience in analyzing large household, facility-based surveys and Family Planning and Reproductive Health data.
- Proven high-level proficiency in statistical analysis and management of large datasets (e.g., Demographic Health Surveys and censuses) using software's like Stata, R, SPSS and Power BI.
- Evidence of scientific publications and scientific conference presentations is required.
- Ability to work and engage with diverse in country stakeholders at national and county level.
- Experience in qualitative data analysis and longitudinal data analysis will be added advantage.
- Excellent report writing and communication skills in English.